

Which model fits this data?

Training-free model-family selection across 47 datasets

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Abstract. Choosing a model for a new dataset normally means training several and comparing them, which costs exactly the effort the choice was meant to save. We ask whether properties measurable *before* training can answer it instead. Across 47 datasets spanning 13 domains, evaluated under one prediction task with eight method families, the architecture is a second-order effect: choosing the best method over the second-best buys a median of $0.038 R^2$, while the achievable ceiling varies by 1.232 across datasets, a factor of 33. Two things follow. First, that ceiling is itself predictable without training: an estimated noise floor correlates with the best achievable R^2 at $\rho = -0.87$. Second, a model-family selector built on seven such properties beats every fixed strategy and, critically, holds up when entire domains are held out (mean regret 0.038 against 0.063 for the best fixed choice, $p = 0.013$). The practical motivation is a finding we did not anticipate: neural networks do not degrade gracefully. On 18 of 47 datasets they land within 0.05 of the best available method; on 16 they fall below $R^2 = -0.5$, worse than predicting the training mean. Which side you land on is predictable from rows-per-dimension and coverage before a single gradient step.

1. Introduction

The standard way to pick a model is to fit several and keep the best. That is defensible when training is cheap and honest when the comparison is fair, but it inverts the usual order of enquiry: we learn about the data by spending compute on models, rather than deciding how much compute the data deserves. This paper asks whether the second order is available.

We frame the question as model-*family* selection. Given a dataset never seen before, and only statistics computable without fitting anything, which family of method should be reached for — a trivial baseline, a classical statistical method, or a neural network? We answer it with a selector validated out of sample, and scored by regret rather than accuracy, since recommending a method that was very nearly as good as the winner is not a real cost.

Our contributions are: (i) a quantification, across 47 datasets and 13 domains under one task, of how much the choice of method matters relative to the choice of dataset; (ii) evidence that the achievable ceiling is predictable from training-free properties; (iii) a model-family selector that transfers across held-out domains; and (iv) the observation that neural networks in this regime either match the best method or fail catastrophically, with little in between, which is what makes advance prediction worth having.

2. Setup

Data. 47 datasets across 13 kinds: weather, sales, macroeconomics, sensors, audio, biomedical panels, chemistry, images, six simulated chaotic systems, and three controls whose answers are known in advance (white noise, a periodic signal, a random walk). The controls calibrate the pipeline: the best of all methods reaches $R^2 = -0.000$ on white noise and $+1.000$ on the periodic signal.

Task. One task everywhere, so nothing is confounded by framing. Standardise, take a window of the last eight observations, predict the next. Train and test are contiguous blocks separated by a gap of 5% of the series, so nothing adjacent in time leaks across the split.

Methods. Eight families spanning what practitioners actually reach for: the training-set mean and persistence (the floors); ridge regression, k-nearest-neighbour analogue lookup, and gradient boosting (classical); and an MLP, a 1-D CNN and a GRU (neural). Three seeds each, 1,128 finite results.

Properties. Seven quantities computed without training: coverage (median distance from a test point to its nearest training neighbour, over the spread of the training set); an estimated noise floor (near-identical inputs whose next values disagree); distribution shift between train and test; decorrelation time; spectral entropy; dimension; and length.

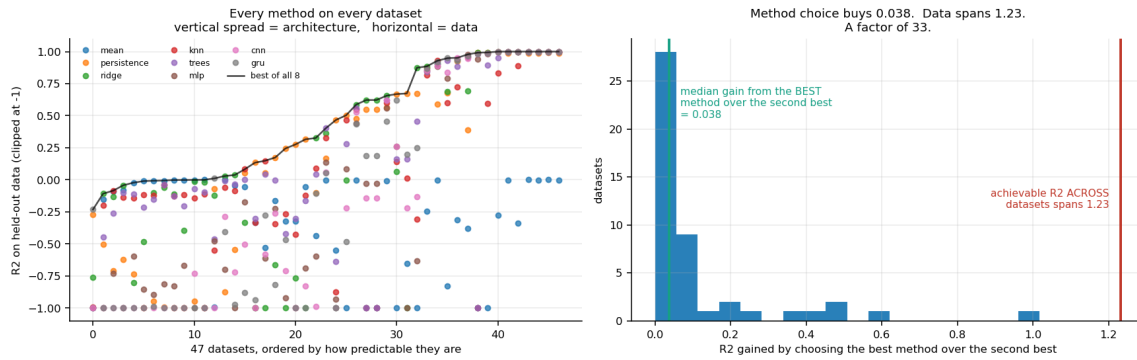


Figure 1: Every method on every dataset, ordered by predictability. Vertical spread is the architecture; horizontal spread is the data. Choosing the best method over the second best buys 0.038 R^2 ; the achievable ceiling spans 1.232.

3. Results

3.1 The choice of method is a second-order effect

Figure 1 shows all eight methods on all 47 datasets. The median gain from picking the best method rather than the second best is 0.038 R^2 ; the best achievable R^2 ranges from -0.232 to $+1.000$, a span of 1.232. The data therefore accounts for roughly 33 times more variation than the method does.

Counting winners makes the same point differently. Ridge regression is the most frequent winner at 14 of 47; persistence wins 10 and the training mean 8, so the trivial baselines together take 18. A neural network wins on 9.

One split must be made before any of this is read as a claim about architectures. Nineteen of the 47 datasets are tabular panels with no genuine time axis, included deliberately to see whether the pipeline would notice. It does: median best R^2 is 0.029 for those against 0.880 for the 28 genuinely temporal datasets, and a neural network wins 1 of 19 against 8 of 28. Asking a sequence question of data with no sequence yields nothing, correctly, and those datasets are excluded from any architectural conclusion.

3.2 Neural networks tie or fail, rarely in between

Across all 47 datasets the best neural network lands within 0.05 R^2 of the best available method on 18, and below $R^2 = -0.5$ — worse than predicting the training mean — on 16. Only 13 sit between. The distribution is bimodal, not graded.

The near-ties are genuinely near. On weekly CO_2 the network reaches 0.984 against ridge's 0.992; on monthly sunspots 0.867 against 0.886. It did not fail; it simply did not beat a linear model, at considerably greater cost. The failures, by contrast, are total: $R^2 = -1.000$ on *ecoli* (335 rows) and -0.931 on *wine* (178 rows), where a trivial baseline reaches 0.276 and 0.467.

Which side a dataset falls on tracks its size. Above 5,000 rows the median gap to the best method is 0.000 and one of nine datasets blows up; below 1,500 rows it is close to a coin flip. Rows per dimension separates the two groups more sharply still: a median of 274 where the network survived against 81 where it inverted. Coverage predicts the gap at $p = +0.62$.

This is the practical motivation for the rest of the paper. A method that degrades gently is safe to try; one that goes from tying the best to worse than the mean, with no warning, is not.

3.3 The achievable ceiling is predictable without training

For each dataset we take the best R^2 achieved by any of the eight methods and correlate it against each training-free property.

Property	Spearman ρ	p
Estimated noise floor	-0.871	< 0.0001
Spectral entropy	-0.739	< 0.0001
Decorrelation time	+0.662	< 0.0001
Coverage	-0.461	0.001
Dimension	-0.403	0.005
Length (rows)	+0.356	0.014
Distribution shift	+0.072	0.63

Table 1: Training-free properties against the best R^2 any method achieved, 47 datasets.

The noise-floor estimate alone accounts for most of it. Decorrelation time is the sharpest categorical signal: a decorrelation time of one step — consecutive rows independent — appears on every dataset where nothing works, regardless of method.

Distribution shift is the exception, and it is informative. It does not predict the ceiling at all ($p = 0.63$). In a separate controlled experiment we added irrelevant columns of four kinds to a fixed dynamical system. Non-stationary columns — those whose distribution moves between train and test by 0.83 of their own standard deviation, against 0.01–0.03 for the others — raised the network's error by a factor of 896 and produced a generalisation gap of 38,415, while leaving a lookup baseline untouched. Non-stationarity does not lower what is achievable; it destroys the model you fitted. These are different failures needing different responses, and only the first is a property of the data.

3.4 A selector that transfers across domains

We fit a random forest to predict which of the three families achieves the best R^2 , using only the seven properties. It is scored by regret — the R^2 lost against the best method actually available — under two validation schemes: leave-one-dataset-out, and the far harsher leave-one-domain-out, in which every dataset of a kind is held out together.

Strategy	Regret (dataset held out)	Regret (domain held out)
Oracle (unreachable)	0.000	0.000
Selector, from data properties	0.040	0.038
Always trivial	0.063	0.063
Always classical (majority)	0.119	0.119
Random choice	0.201	0.201
Always neural	0.419	0.419

Table 2: Mean regret, lower is better. The selector beats the best fixed strategy by 1.6 $\times$ and always-neural by 11 $\times$.

The result that matters is that domain-holdout regret (0.038) is no worse than dataset-holdout regret (0.040). This was our pre-registered failure condition: had the selector only worked because it had seen four other chaotic systems, it would have learned the collection rather than a property of data. Paired Wilcoxon tests against the fixed strategies give $p = 0.013$ (always trivial), 0.006 (always classical) and < 0.0001 (always neural).

Family accuracy is 60% against a 33% chance baseline, which is modest; regret is the honest metric and is where the result lives. The winning method's identity also flips across seeds on 21 of 47 datasets, so this is a regime recommendation rather than a per-dataset oracle.

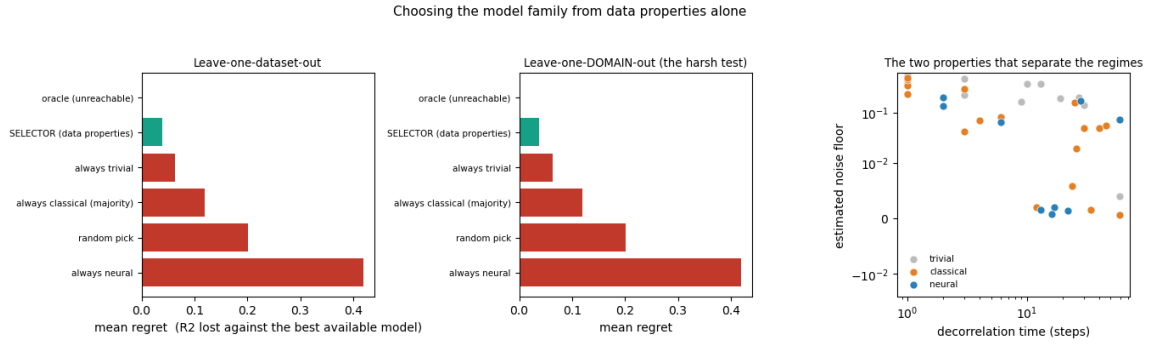


Figure 2: Regret against fixed strategies. Middle panel is the leave-one-domain-out test. Right panel shows the two properties that separate the regimes most cleanly.

The learned rule is legible. A depth-3 tree splits first on the noise floor, then on coverage, then on spectral entropy or distribution shift. Feature importances are spread — coverage 0.21, noise floor 0.18, dimension 0.14, rows 0.14, spectral entropy 0.14, shift 0.11, decorrelation 0.09 — which is why single correlations look weak while the combination does not. The regimes have distinct profiles: where a neural network wins, the median decorrelation time is 16 steps and coverage 0.077; where a trivial baseline wins, 2 steps and 0.747.

4. Limitations

Scale. Every result was measured at 1–60 dimensions, 150–42,000 rows and 10^4 – 10^5 parameters, on one prediction task with a fixed window. The claim that architecture is second-order is measured only where every method is small, and is the claim most likely to invert at scale.

The noise-floor estimator degrades with dimension. Against synthetic data with a known ceiling of 0.80 it reads 0.80 at four dimensions, 0.68 at eight and 0.42 at 32, always pessimistically, because it requires near-identical inputs and in high dimensions nothing is near anything. Above roughly 16 dimensions it should be read as a lower bound. This is our most useful single result and its most exposed one.

Selector size. The selector is fitted on 47 points. Domain holdout is the strongest available test at that size, but it is still 47.

Not tested. Multi-task, action-conditioned and representation-learning settings. A model whose predictions influence the data it later sees is outside everything measured here.

5. Related work

That simple baselines often match or beat deep models on time series is well established, from the M-competitions through to linear models outperforming transformers on standard forecasting benchmarks. Our contribution is not that observation but the question downstream of it: *which measurable property of the data decides it*, and does that property transfer to a domain the predictor has never seen. Analogue forecasting, and the classical result that natural analogues become vanishingly rare as dimension grows, supplies the mechanism behind our coverage measure. Work on dataset difficulty and learning-curve extrapolation shares our target but generally requires partial training; the properties here require none.

6. Conclusion

On this evidence the dominant practical decision is not which network to build but whether to build one, and that question is answerable in seconds from the data. The ceiling is predictable at $\rho = -0.87$ without training; family selection beats every fixed strategy and survives domain holdout; and the reason it is worth doing is that neural networks in this regime fail catastrophically rather than gracefully.

Two recommendations follow for practice. Report a training-free baseline alongside any surrogate result — across 47 datasets a trivial or classical method won 38 times. And report the data regime alongside the error, because two results with the same R^2 in different regimes have not demonstrated the same thing.

Reproducibility

All code, raw per-run results and figures are in the project repository. Every experiment writes one JSONL record per unit of work carrying its configuration hash and git commit, so any number here traces to a run. Datasets are fetched from public mirrors and cached; failures are reported rather than dropped, so a stated dataset count matches what ran. Total compute for the study reported here is 80 minutes on two CPU cores.

AI involvement

This work was carried out with an AI assistant (Claude) acting as the research agent under human direction. The assistant wrote the experiment code, ran the experiments, produced the figures and drafted this paper. The human author supplied the problem statement, chose the research direction at each decision point, supplied the reference material, and directed several changes of course — including the reframing from architecture comparison to model selection that this paper reports. All numerical claims were produced by code in the repository rather than generated by the model. Seven measurement artefacts were found and corrected during the work, four of them in our own instruments; each correction is recorded in the repository history alongside the result it changed.